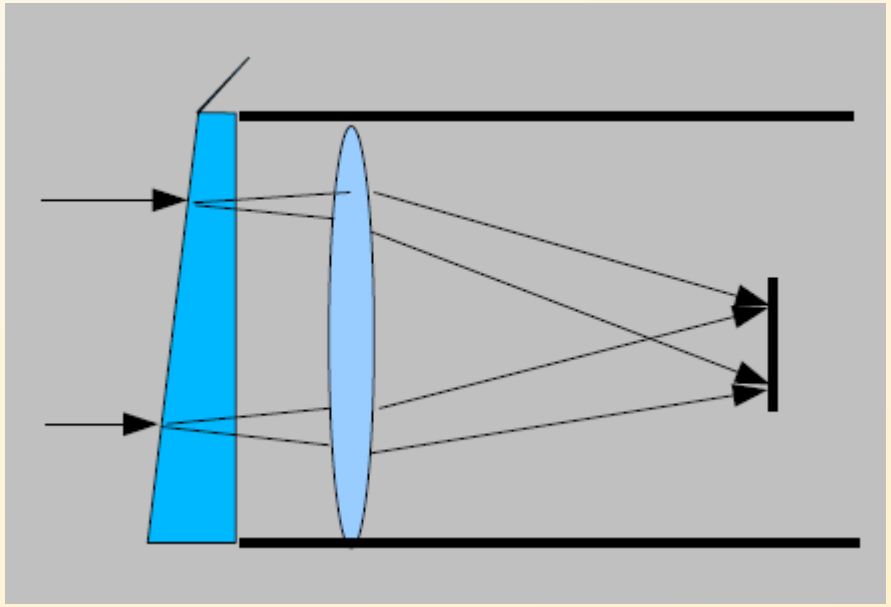
Spectrocope-types: objective prism

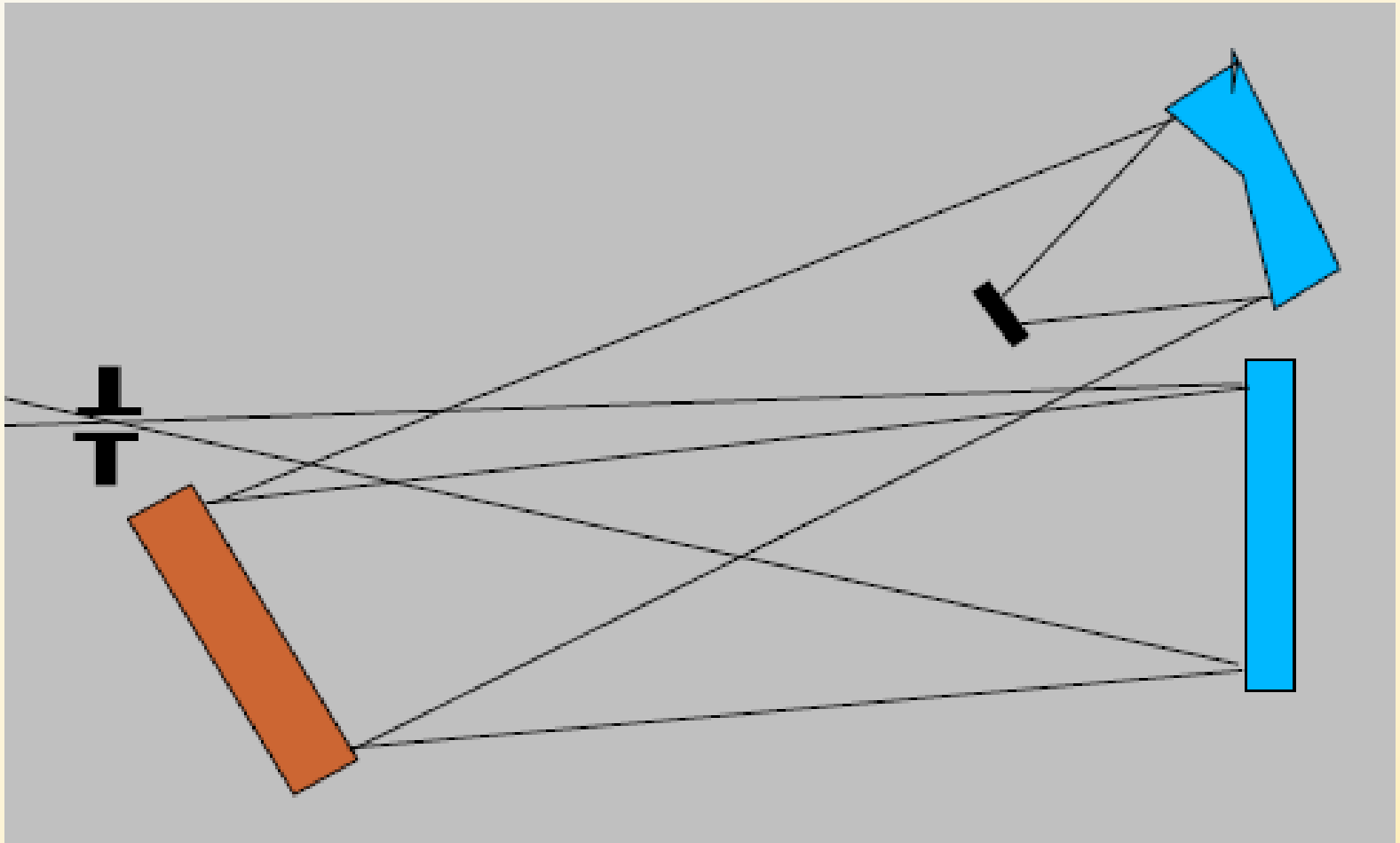
- The simplest form of astronomical spectroscopy is objective prism spectroscopy.
- This consists of putting a large prism in front of a telescope and looking at the sky through that.
- Low resolution and non-linear dispersion.





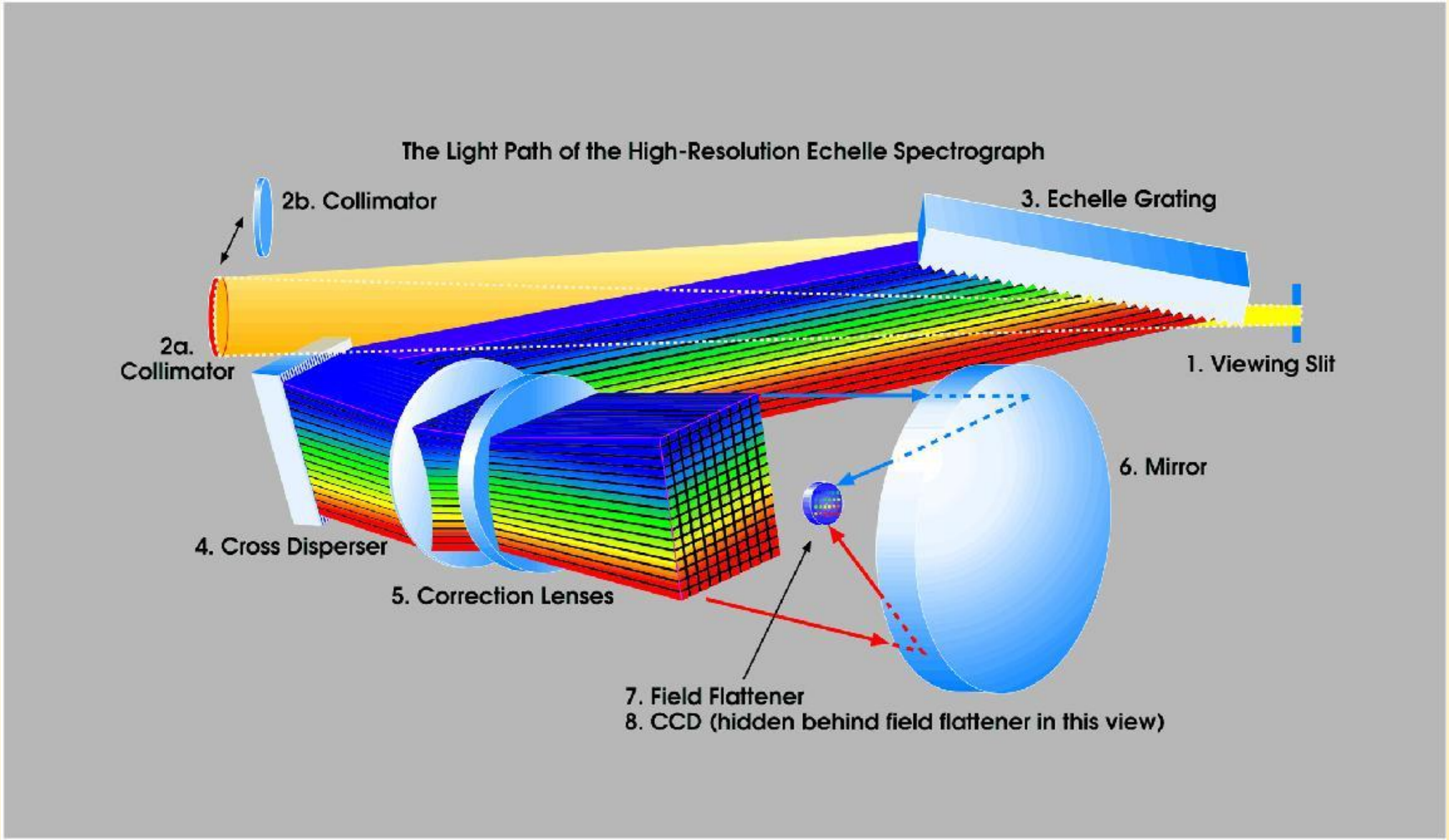
Spectrocope-types: Cassegrain spectrograph

- Produces usually one order.
- Low resolution from R = 1000 to 15,000.
- Almost linear dispersion.



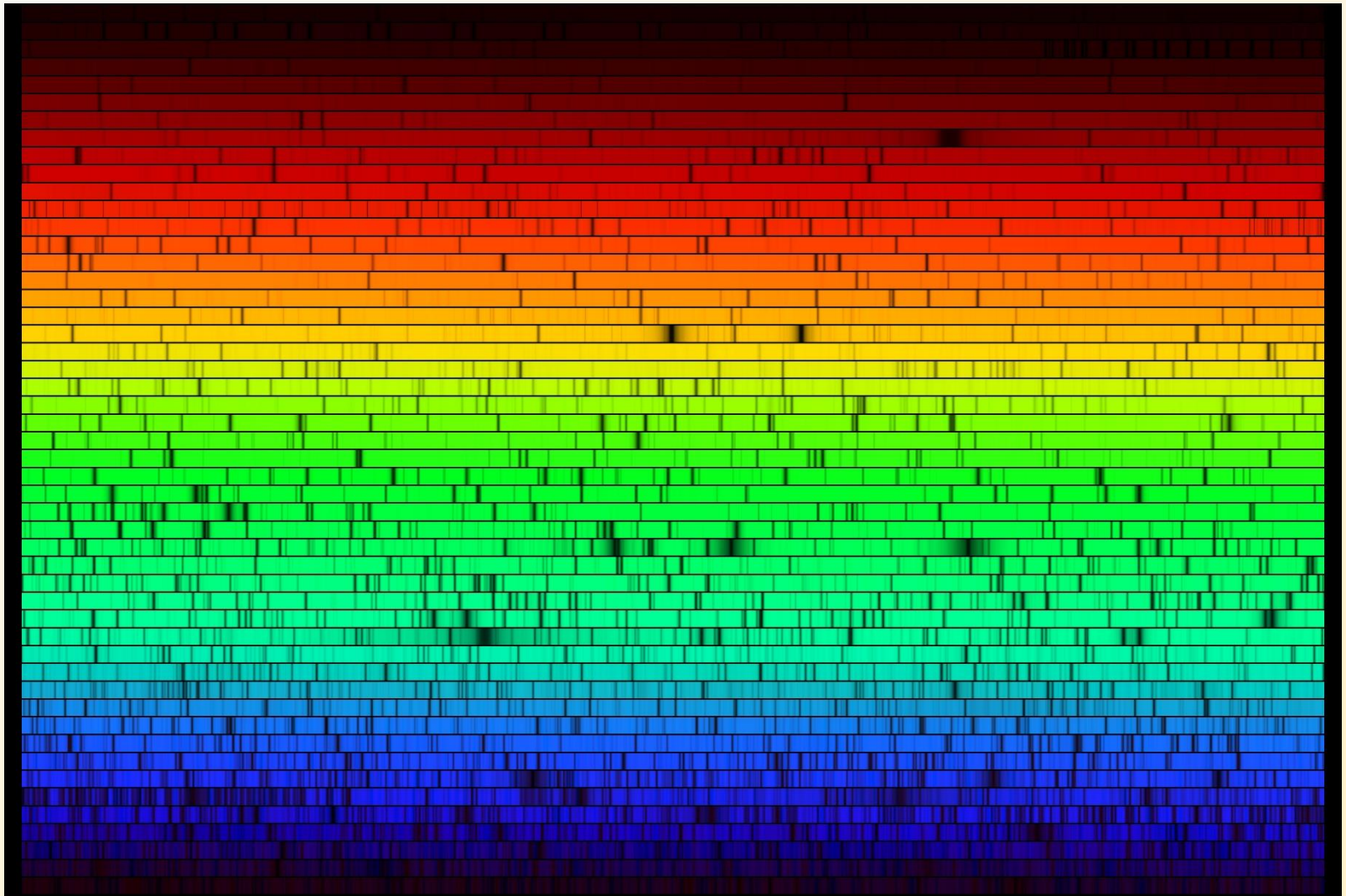
Spectrocope-types: Echelle spectrograph

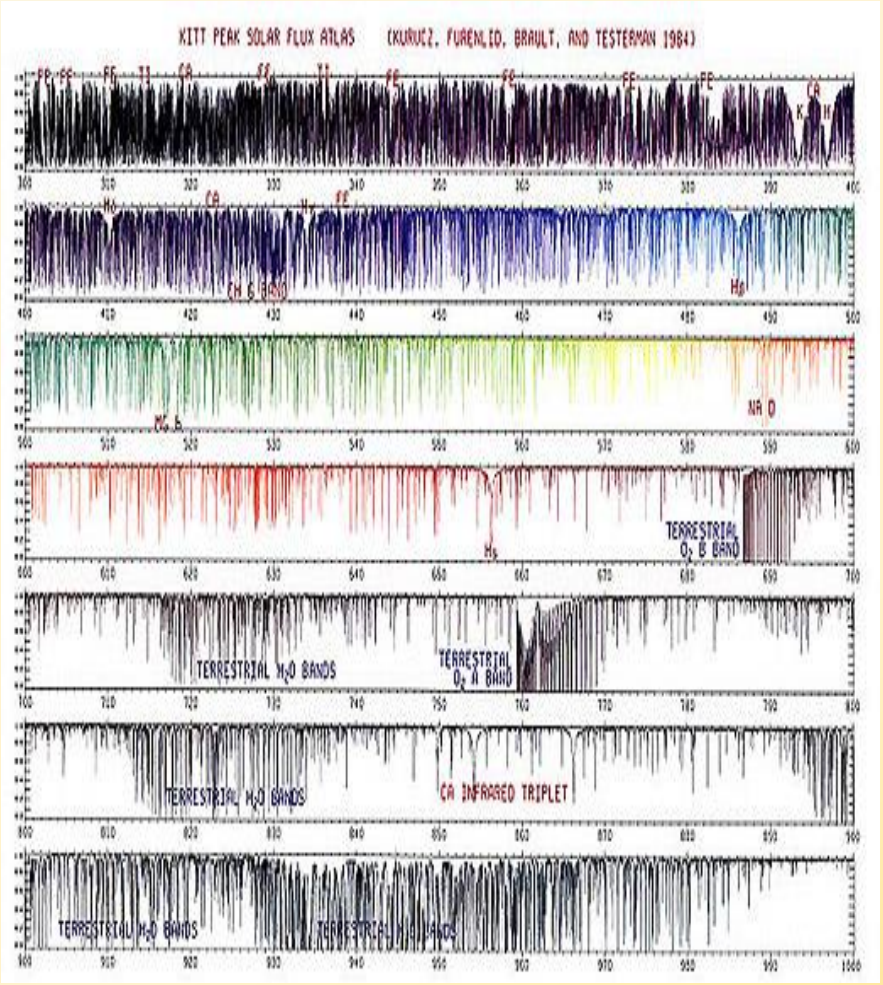
- Splits the spectra into multiple order as a function of wavelength (m > 20).
- High resolution from R = 10,000 to 300,000.



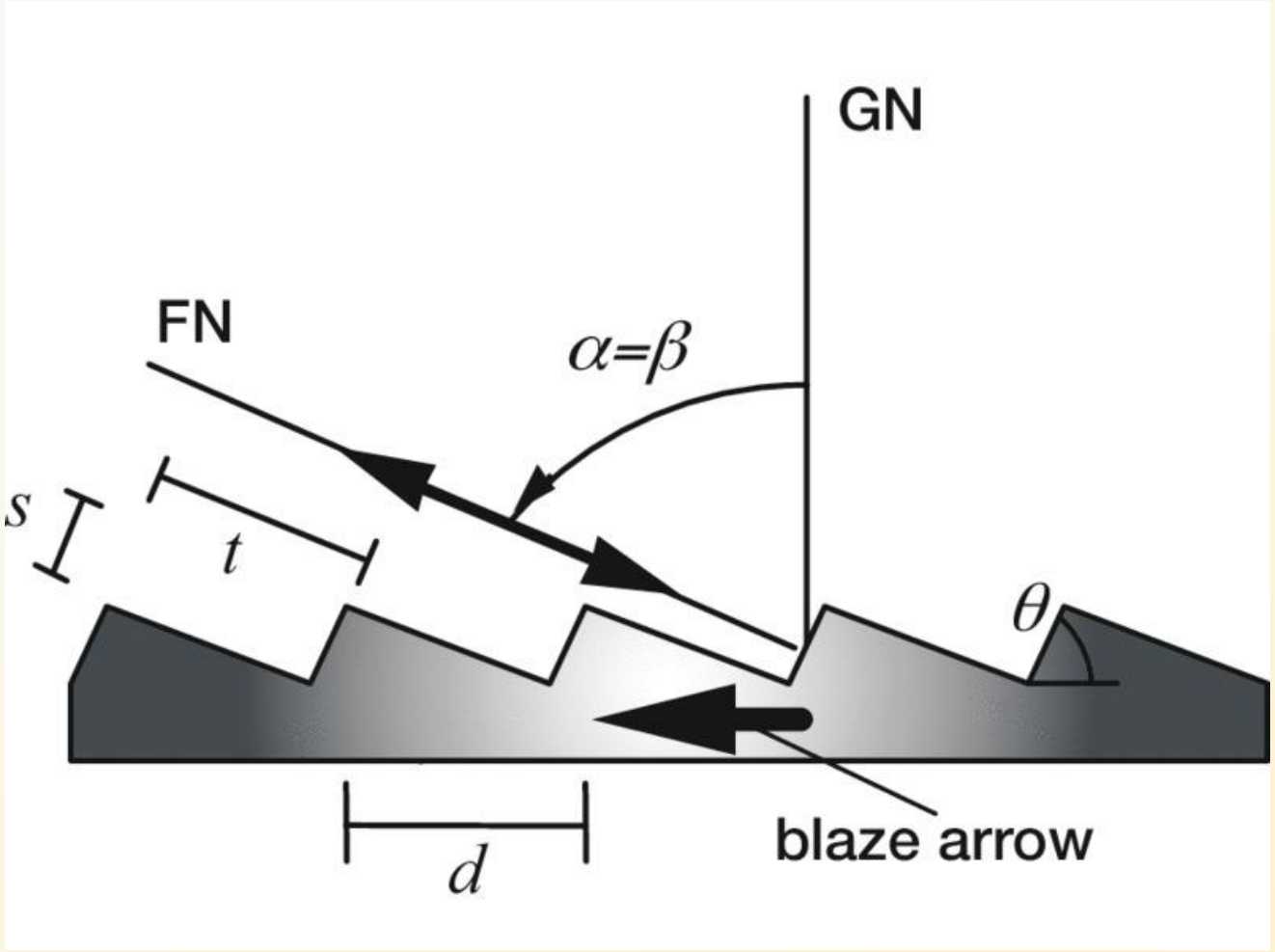
Spectrocope-types: Echelle spectrograph

- Spectrum of the Sun in 50 orders from 400 nm to 700 nm taken with the Fourier Transform Spectrometer at the McMath-Pierce Solar Facility.

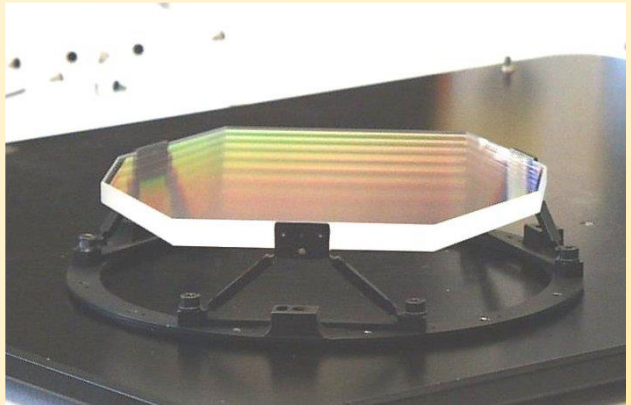




Spectrocope-types: Echelle spectrograph

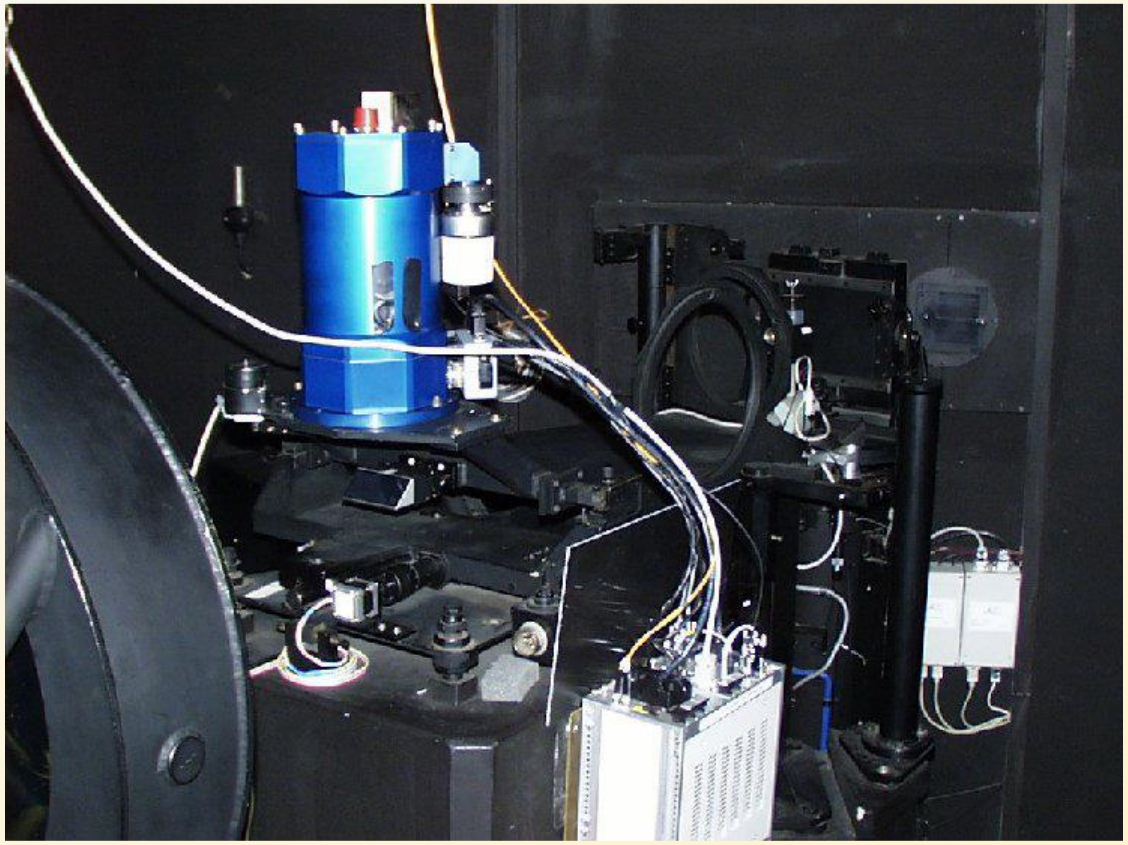


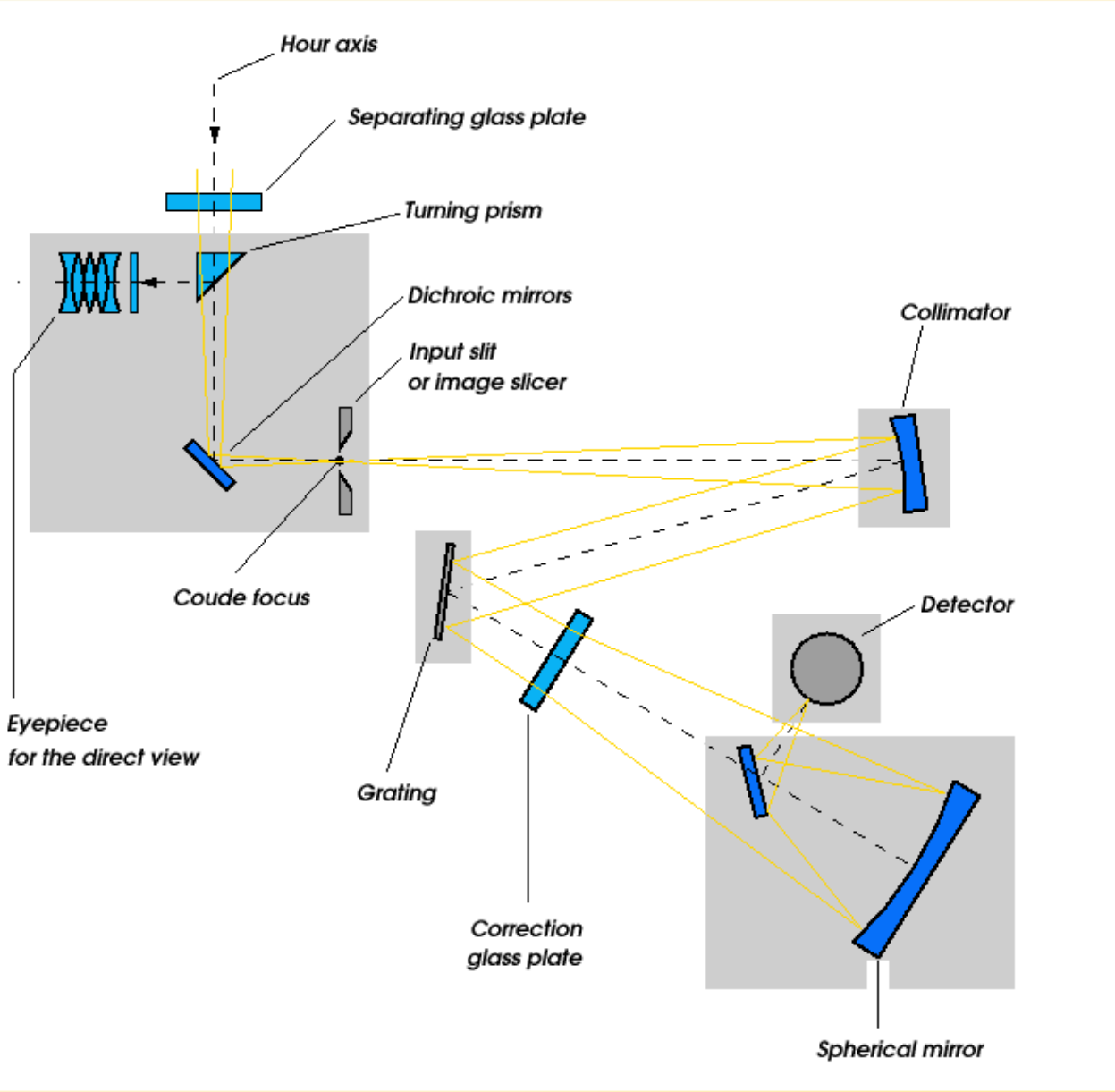
- θ: blaze angle
- d: grating constant.
- Littrow configuration: the angle of incidence α equals the angle of diffraction β.
- Grism: a prism and a grating added together.
- Small size and low resolution.
- Used in IR.



Spectrocope-types: Coudé spectrograph

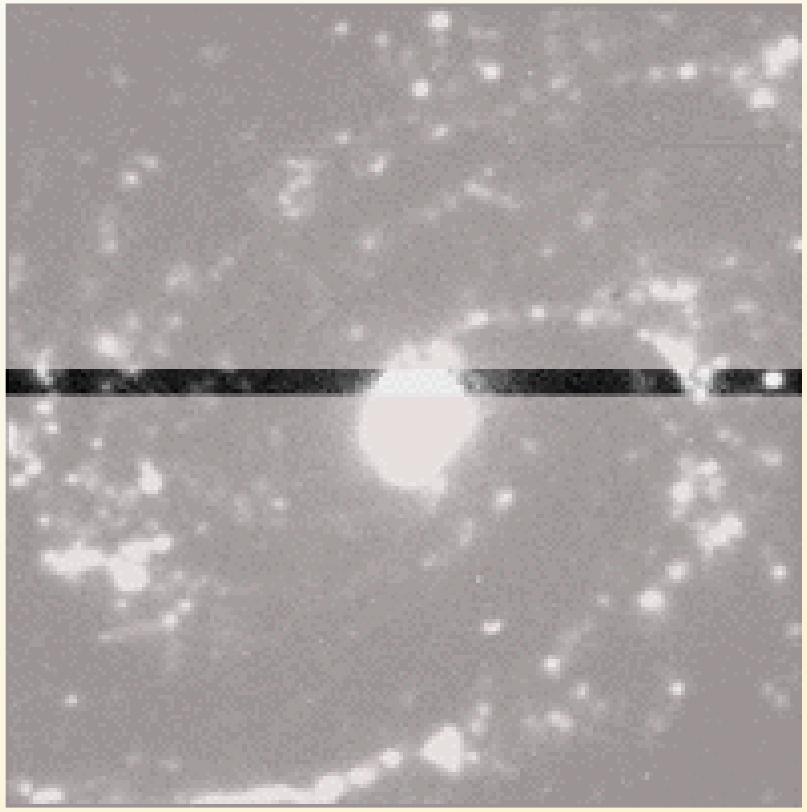
- Spectrograph put into the Coudé focus.
- Large resolution can be achieved using echelle gratings.
- Multiple types can be used.

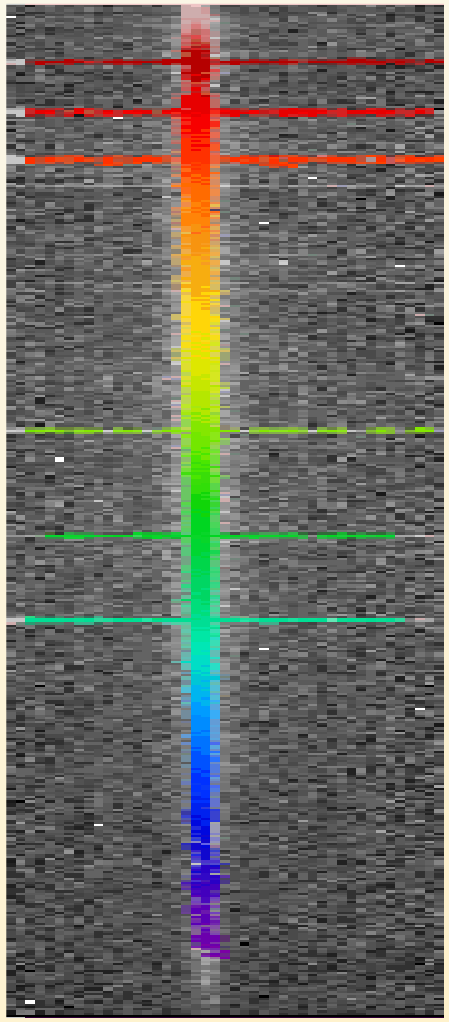


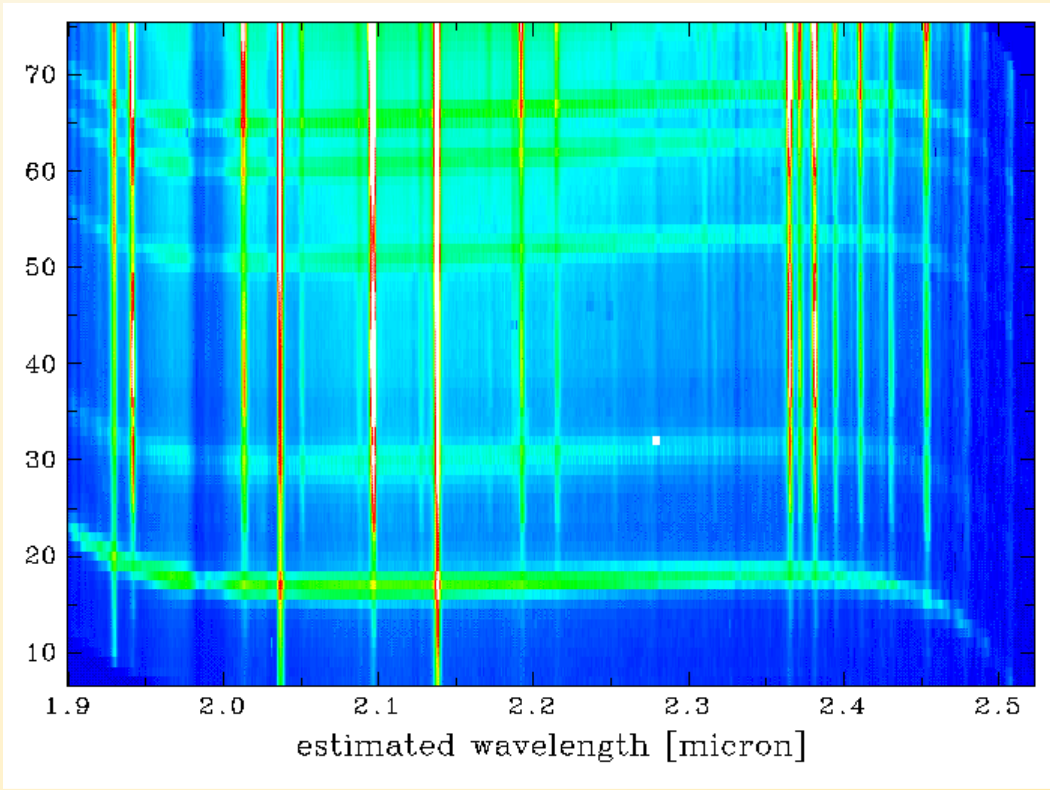


Spectrocope-types: long-slit spectrograph

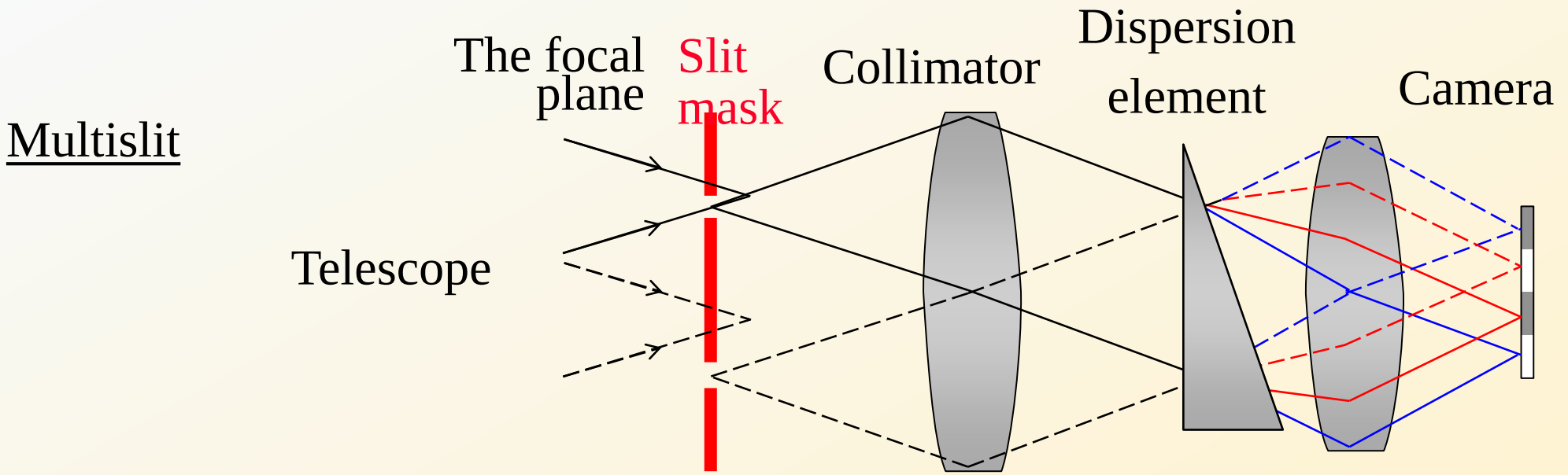
- To examine large objects. It allows to take spectra of multiple parts of the object (and/or multiple objects) along the split.

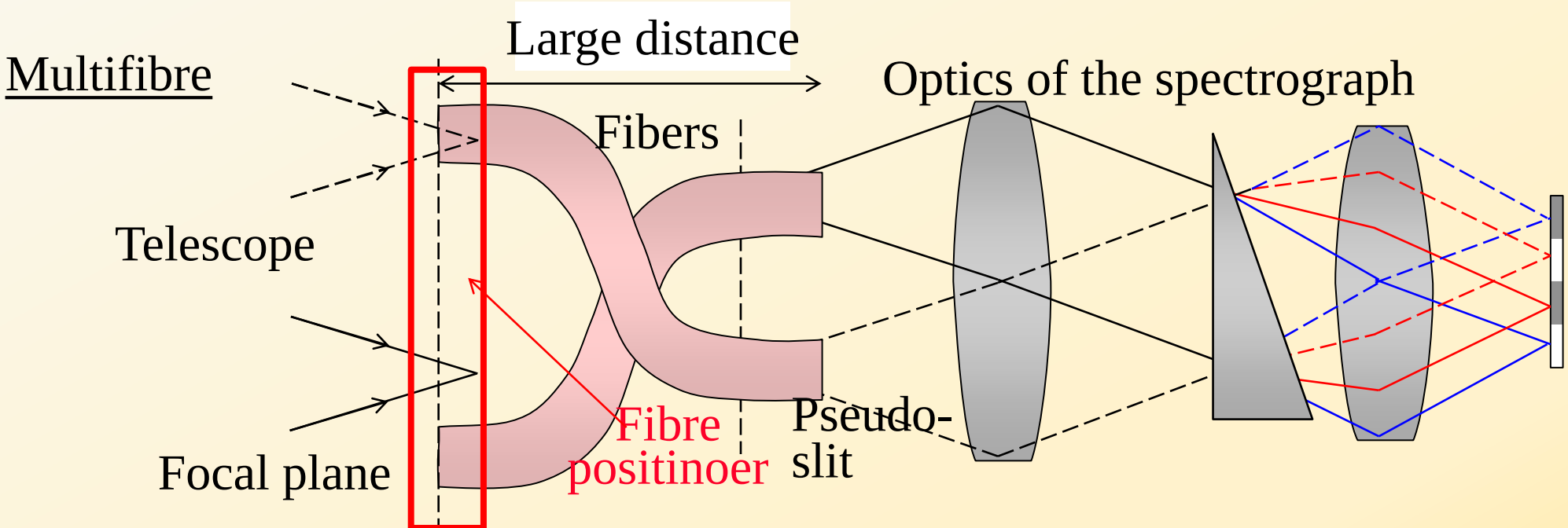




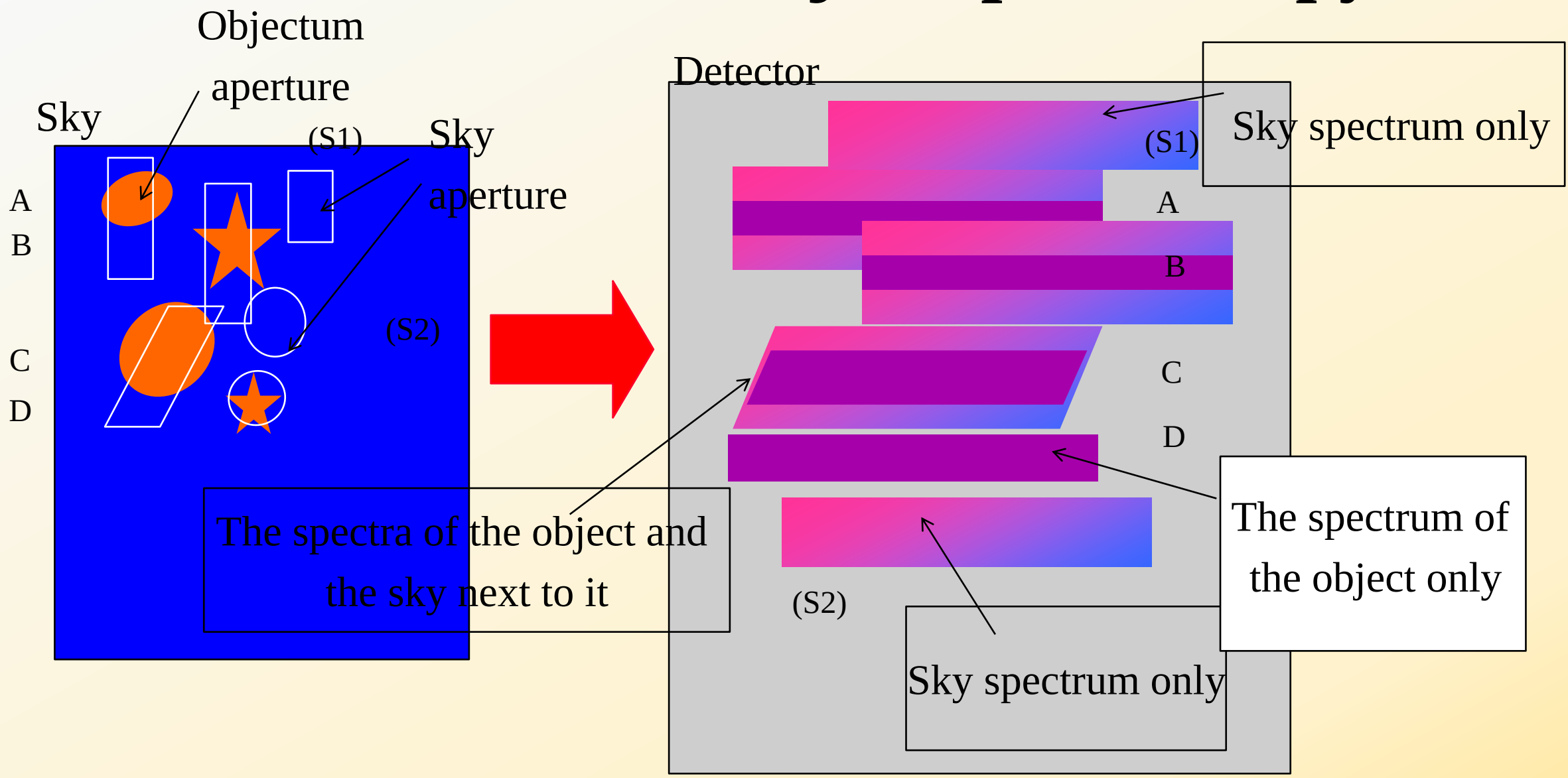


General structure of multi-object spectrograph optics

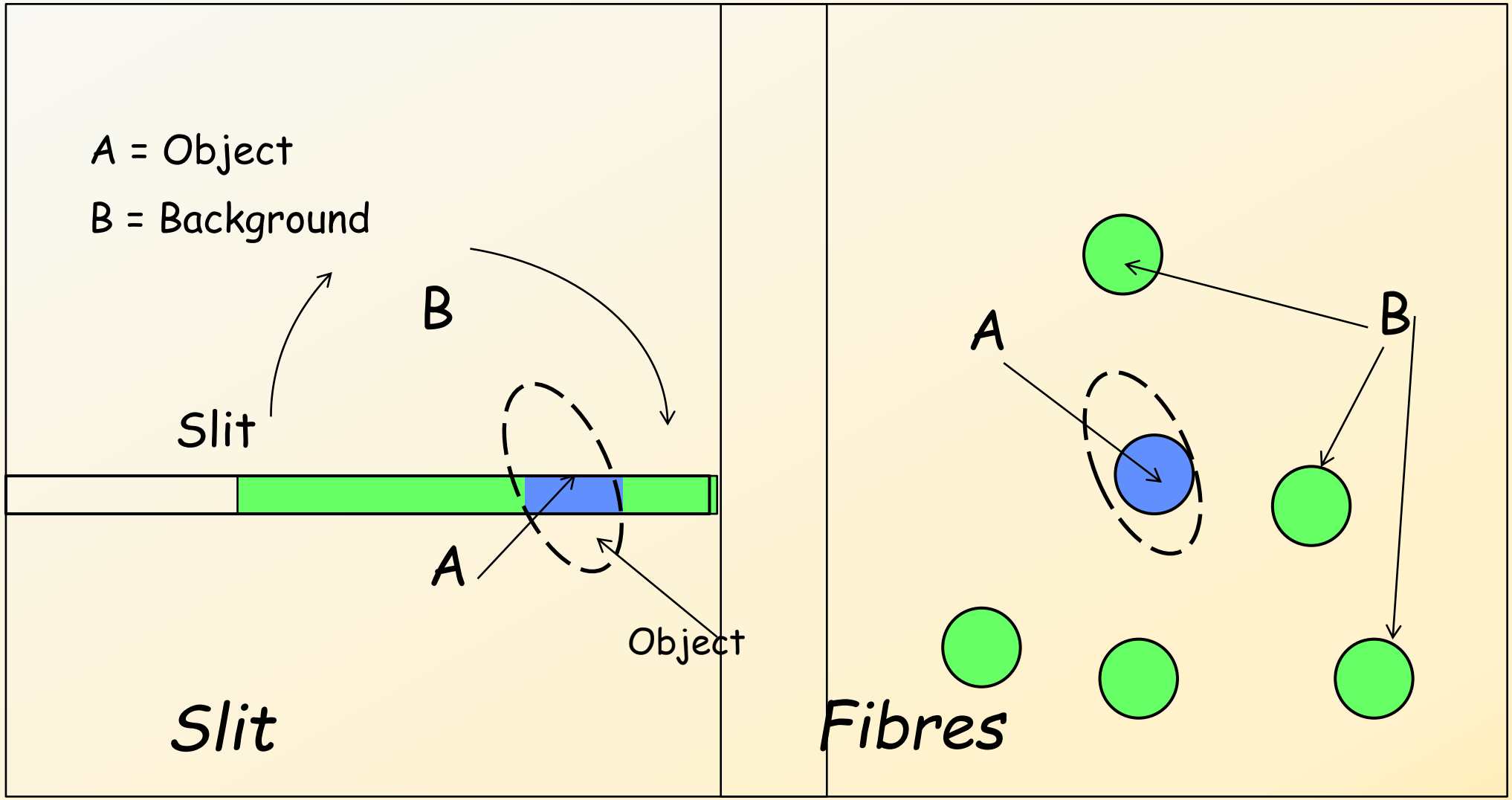




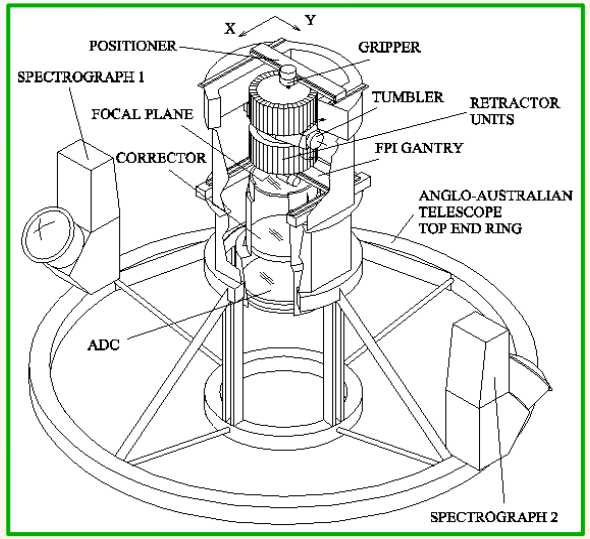
Basic rules of multi-object spectroscopy

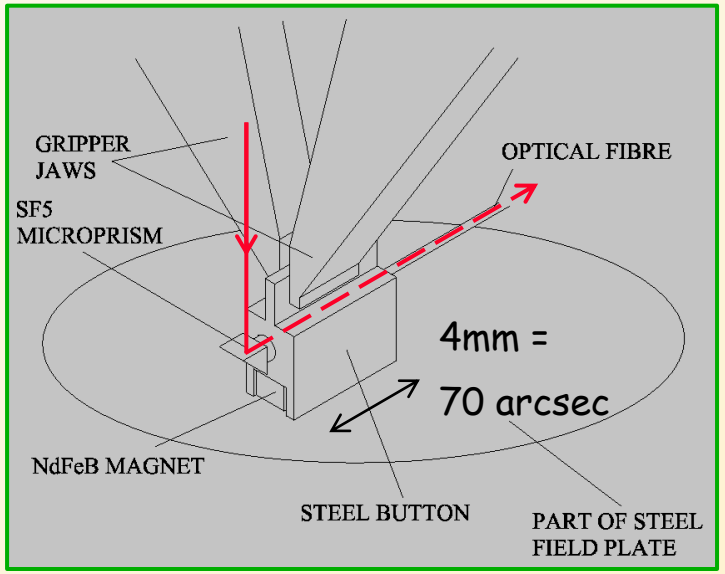


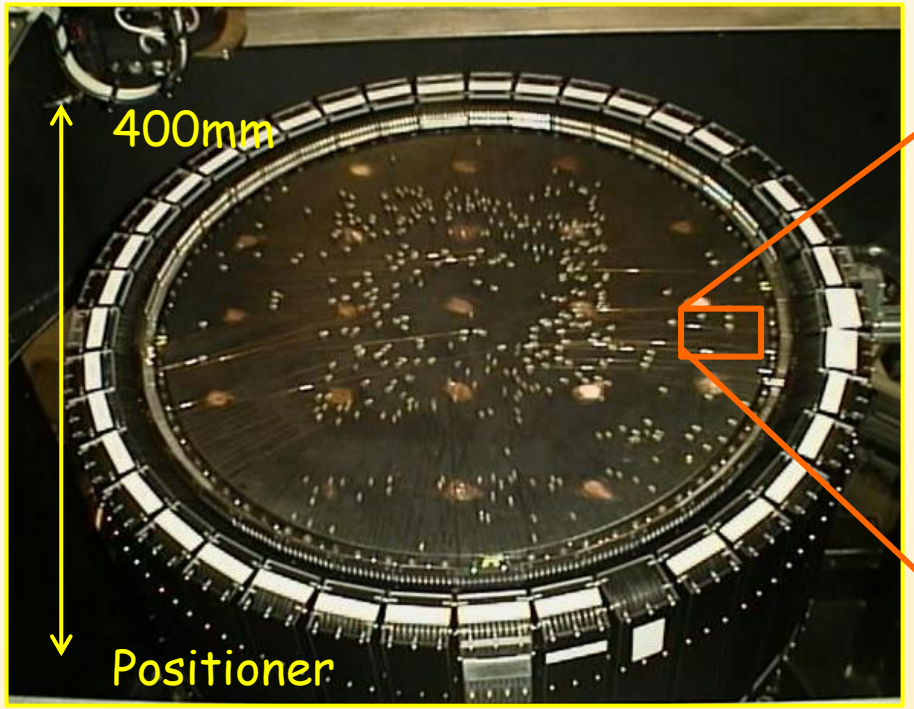
Background subtraction



2df (Anglo-Australian Telescope)



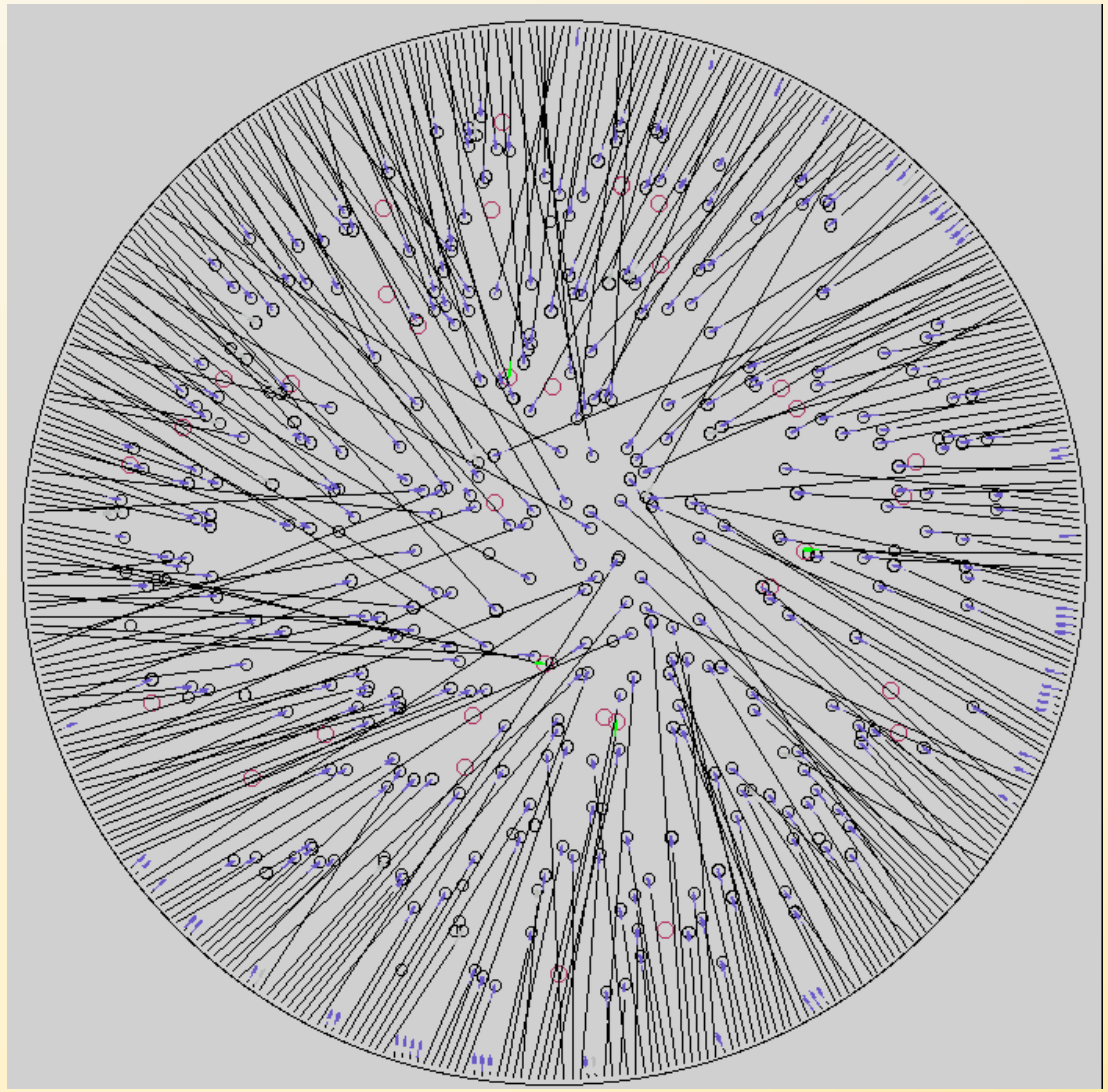






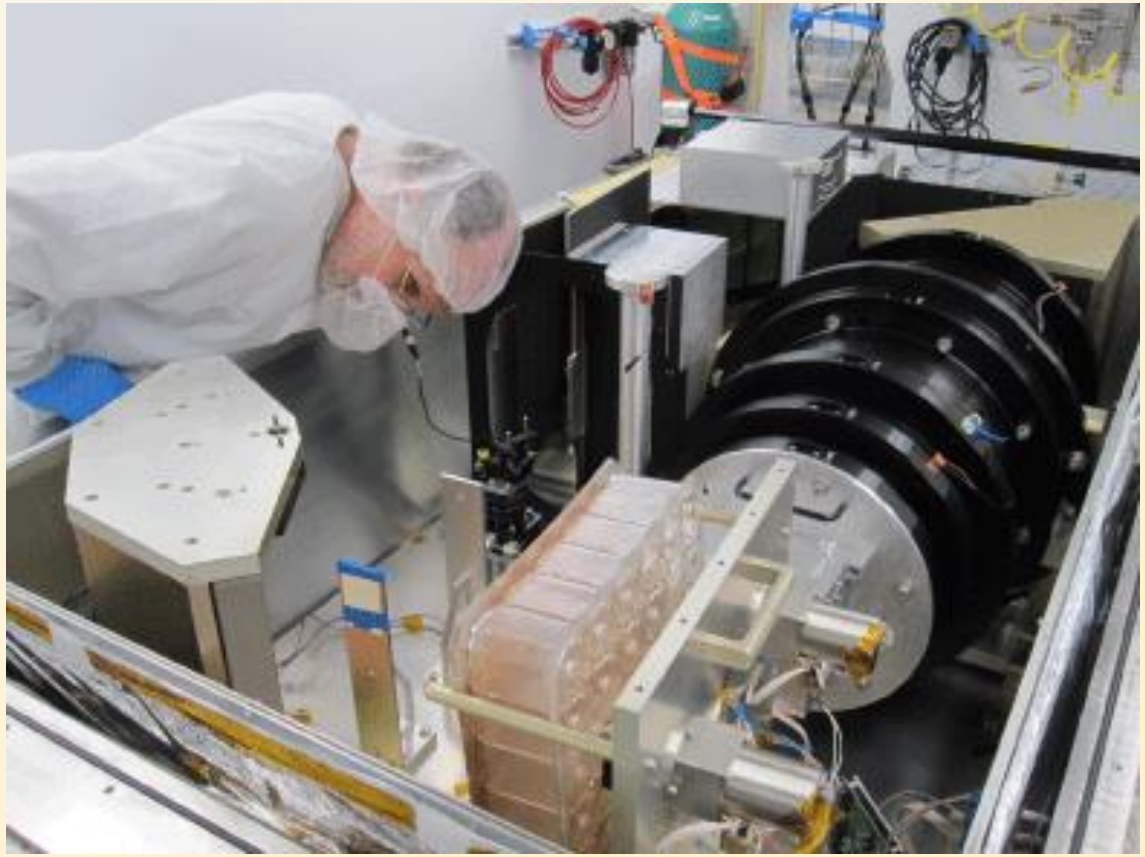
Positioning



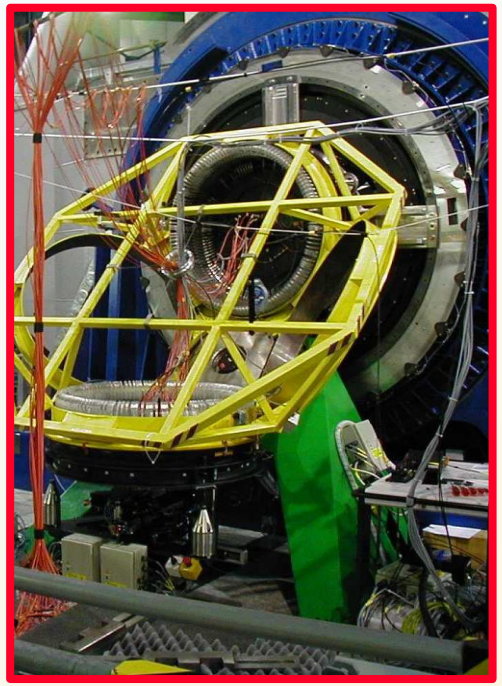


APOGEE spectrograph

- Spectral resolution: 22,500 (approx.)
- Wavelength coverage: 1.51 – 1.70 μm.
- Fiber diameter: 2 arcsec (Northern Spectrograph), 1.31 arcsec (Southern Spectrograph).
- Sensitivity: S/N~100/pixel for H < 12.2 and 3-hour integration.
- Detectors: three JWST H2RG (2048 x 2048) Near-Infrared HgCdTe Detectors with 18 micron pixels.



Flames (ESO VLT)

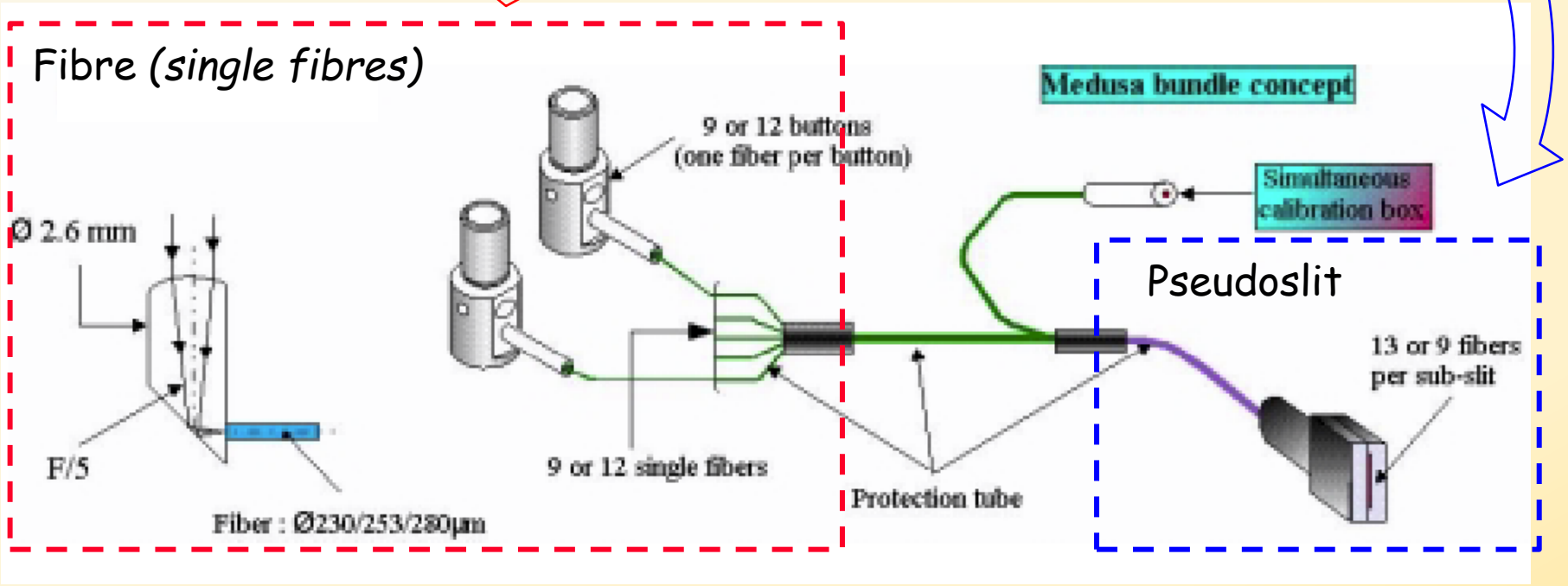


OzPoz (AAO)

fibre positioner at VLT Nasmyth

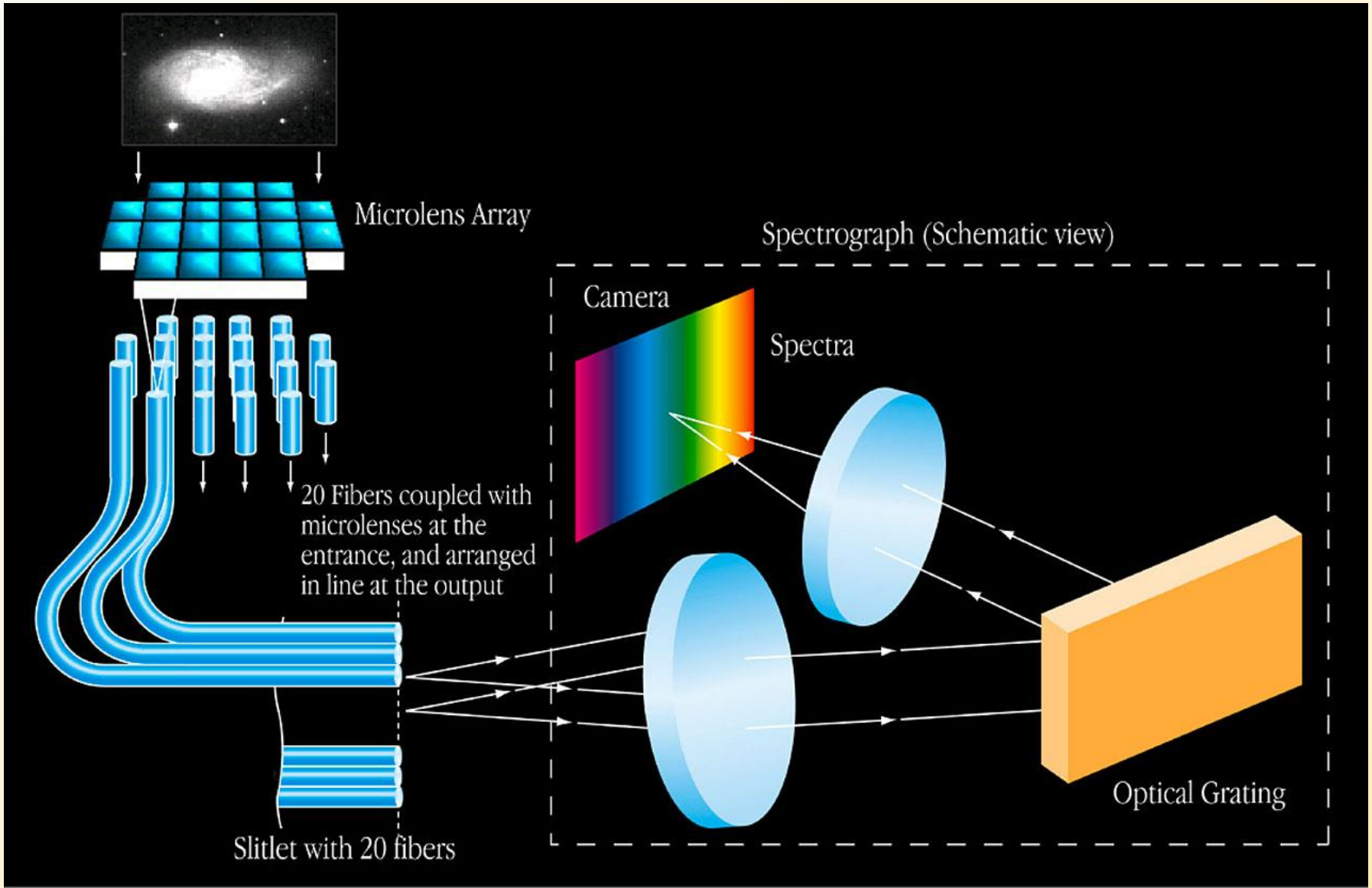
- 0.1" accuracy
- 10" minimum dist.





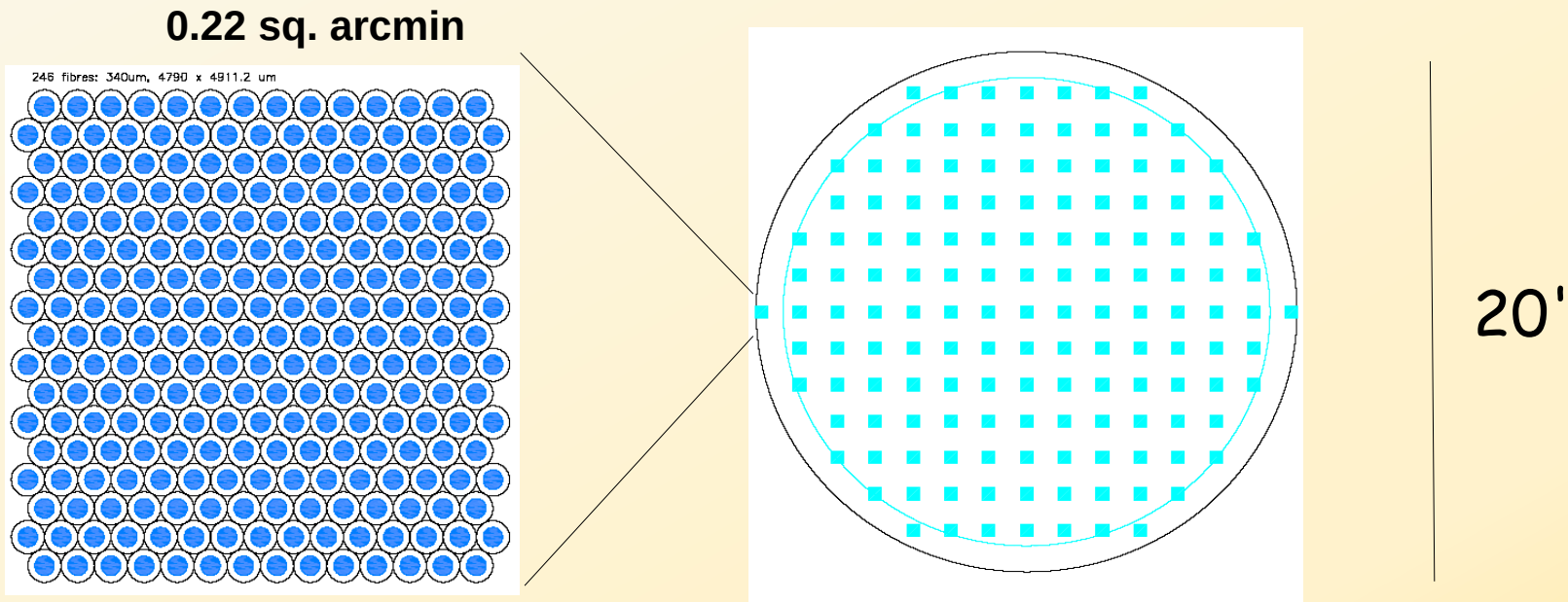
Integrated field units (IFU)

- Multiple fibres in one package.



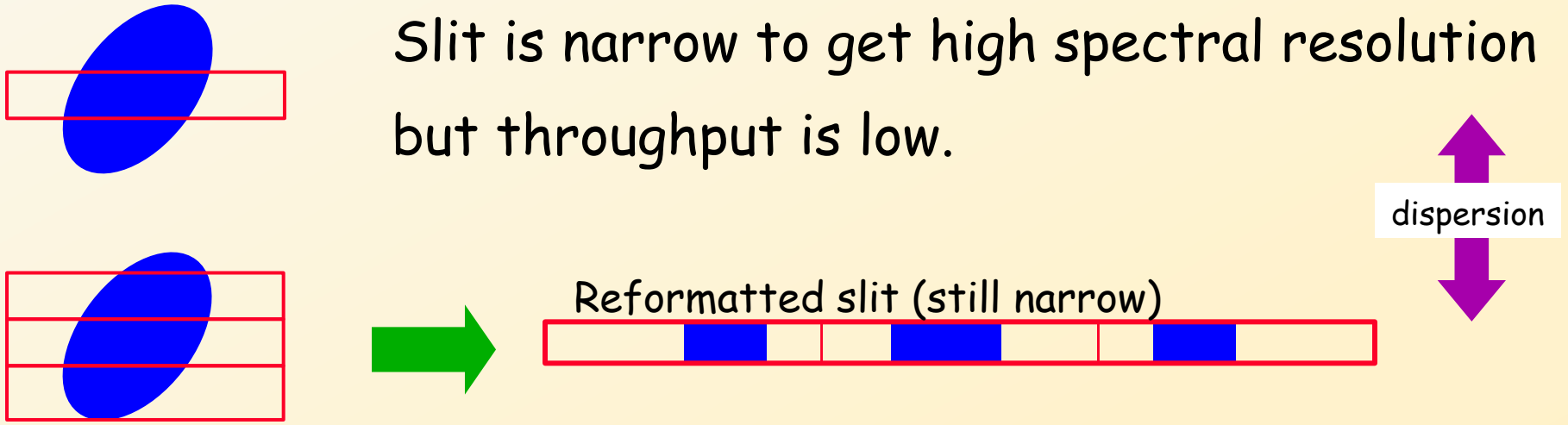
VIRUS (Hobby-Eberley Tel.)

- Visible IFU Replicable Unit Spectrograph.
- 1 IFU: 0.22 sq.arcmin, 340-550 nm, R=850.
- 246 fiber, 1 fiber = 1 sq.arcsec.
- 145 IFU = 30 sq.arcmin per obs.



Slicing

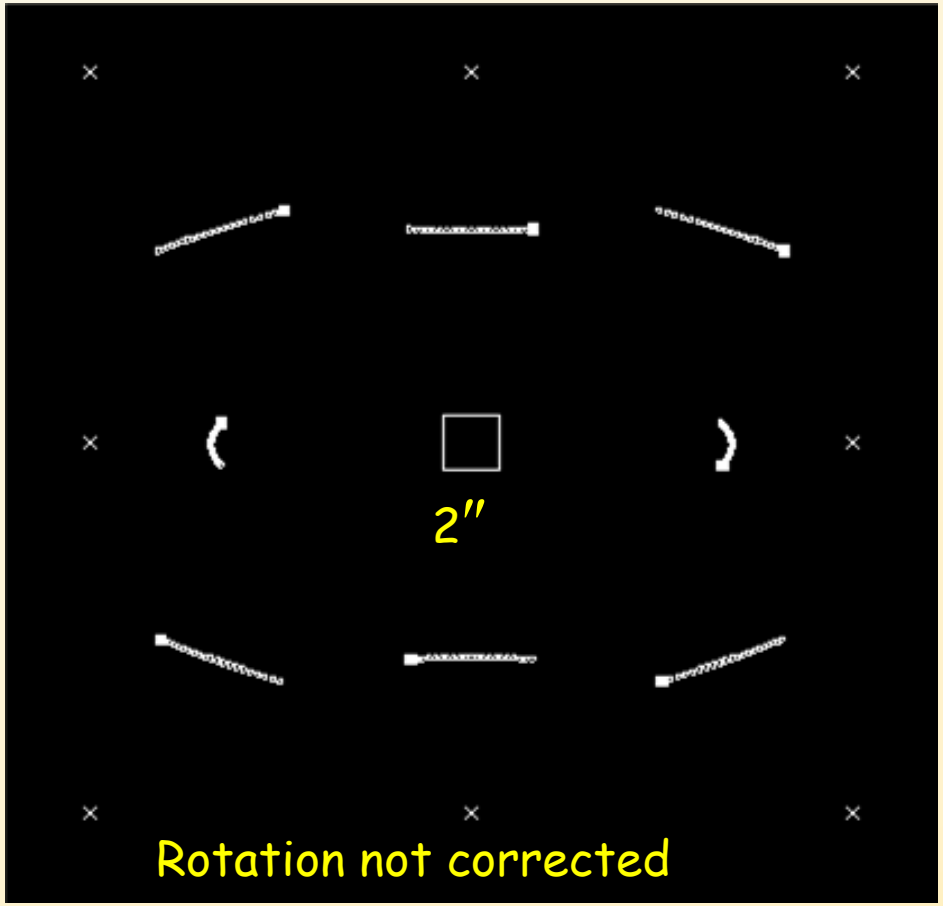
- Gets both high spectral resolution and high throughput.
- Can use fibres (for MOS) or little mirrors.
- Don't confuse with integral field spectroscopy, which retains spatial information.



Slicing allows more light to be captured without sacrificing spectral resolution.

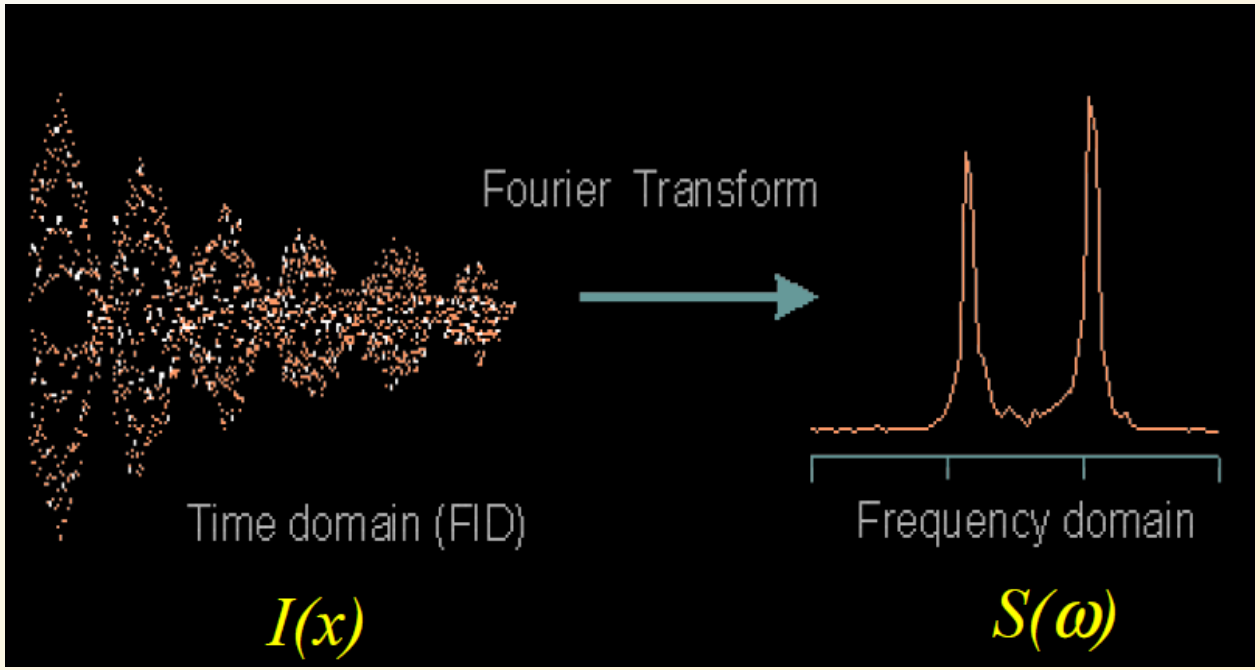
The effects of the atmosphere

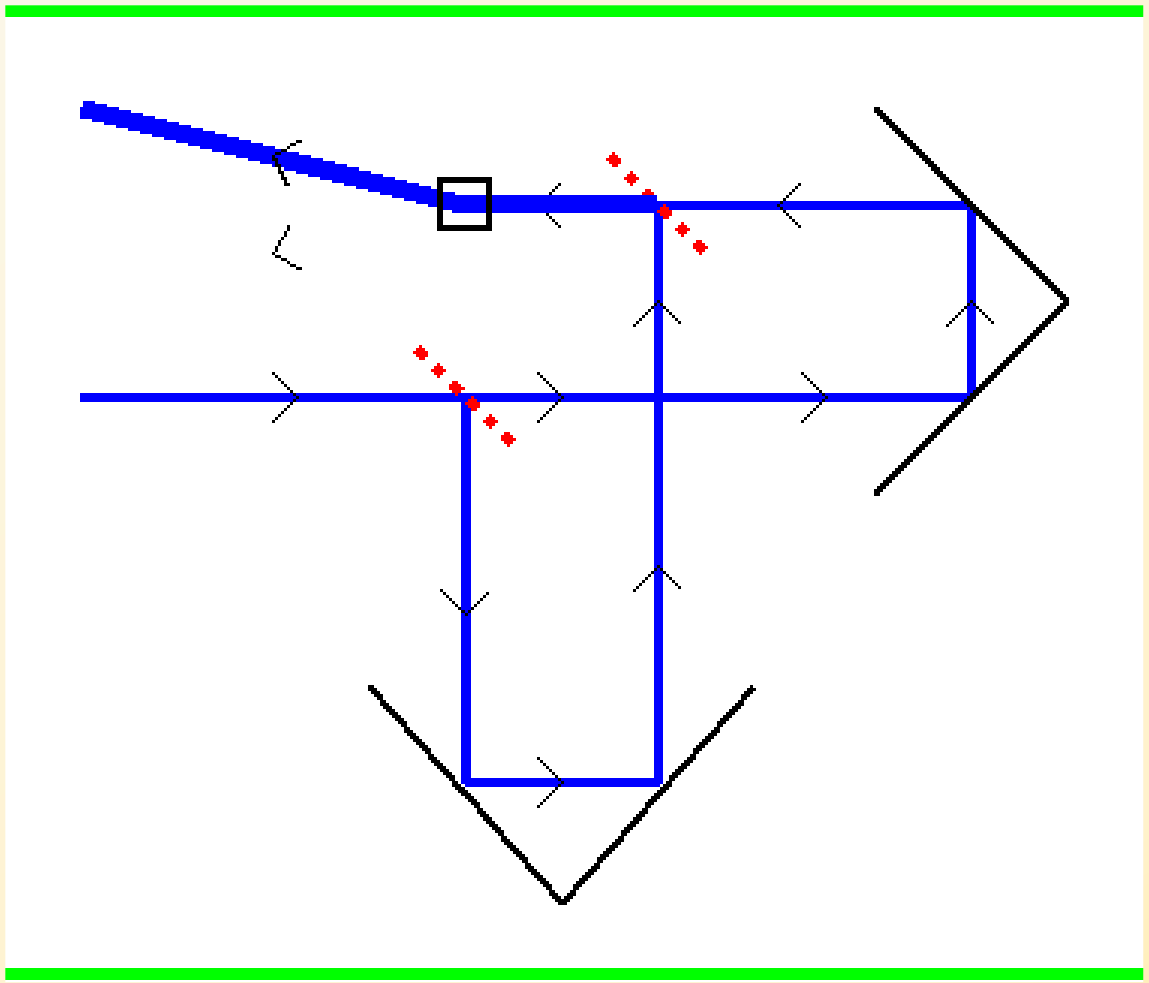
- Parallel-plane model of atmosphere gives error in apparent position of object ("refraction").
- Significant for wide fields and fine image scales.
- Without correction, field is squashed and rotated.
- This also has a chromatic component since refractive index depends on wavelength ("dispersion").
- The correction can be empirically determined by observing iodine cells.



Fourier transform spectroscopy

- The Michelson interferometer uses moving mirrors to obtain an interference image.
- The image changes as the mirror moves.





$$S(\omega) = \int I(x) e^{i\omega x} dx$$

Observations overview

- Taking bias images at the beginning of the night with 0 exposure time to understand the readout noise of the detector.
- Taking dark images with the same exposure of the science images to understand thermal noise. Lower exposure is possible if scaling is good.
- Taking flat images to understand the pixel-to-pixel variations.
- Taking wavelength calibration spectra preferably before and after the science exposures.
- Taking sky images and/or telluric standards to remove atmospheric absorption/emission.
- Taking flux standard/radial velocity standard spectrum if it is needed.
- Taking science spectra.